M&S Monthly Colloquium Meeting

Reliability for multilevel data: A correlation approach

PhD candidate: Tzu-Yao Lin[1,2]

Supervisors: Dr. Francis Tuerlinckx[2], Dr. Sophie Vanbelle[1]

[1]Department of Methodology and Statistics, Care and Public Health Research Institute, Maastricht University, the Netherlands

[2]Quantitative Psychology and Individual Differences Research Group, Faculty of Psychology and Educational Sciences, KU Leuven, Belgium

Maastricht University KU LEUVEN

20-06-2023

Outline

Introduction

Generalizability Theory and the Correlation Approach

Reliability for Multilevel Data

Outline

Introduction

Generalizability Theory and the Correlation Approach

Reliability for Multilevel Data

Reliability and Measurement Error

Researchers in many fields have become increasingly aware of the problem of errors in measurements.

- A **reliable** method or measurement instrument is one which gives the same or similar results when it is repeated.

- Unreliable ⇒ lacking consistency
  - spent a great deal of effort attempting to decipher the different/conflicting results
  - appropriate statistical methods must be employed to enable the correct conclusions to be drawn from the observations.

Table 3 IRR estimates for 20 coefficients on 4 datasets

Coefficient	Diagnoses	Vision	Video	Anxiety
Nominal level				
κ	[a]	0.60*	[a]	[a]
κ_W	[a]	0.65**	[a]	[a]
κ_{W^2}	[a]	0.60*	[a]	[a]
κ_{Fleiss}	**0.43***	0.60*	0.04	−0.04
κ_{Exact}	**0.44***	0.60*	0.10	−0.02
κ_{Light}	**0.46***	0.60*	0.07	−0.02
PA_N	**0.17**	[b]	[b]	[b]
α_N	**0.43***	[b]	[b]	[b]
ι_N	**0.44***	[b]	[b]	[b]
Ordinal level				
W	[c]	**0.85*****	0.39	0.54*
$\bar{\rho}$	[c]	**0.71****	0.24	0.34
PA_O	[c]	**0.71****	[b]	[b]
α_O	[c]	**0.71****	[b]	[b]
Interval level				
PA_I	[c]	[b]	**0.35**	**0**
α_I	[c]	[b]	**0.10**	**0.16**
ι_I	[c]	0.60*	**0.15**	**0.19**
$Finn_2$	[c]	0.78**	**0.92*****	**0.50***
ICC_2	[c]	0.70**	**0.16**	**0.20**
$\bar{r}$	[c]	0.70**	**0.24**	**0.28**
A	[c]	0.85***	**0.40**	**0.48***
Ranges of values				
Range[d]	**0.17 – 0.46**	**0.71 – 0.85**	**0.10 – 0.92**	**0.00 – 0.50**
Range[e]	0.17 – 0.46	0.60 – 0.85	0.04 – 0.92	−0.04 – 0.54

Usefulness of Reliability

- The is an abound of reliability coefficients available
  - depending on levels of measurement
- However, the coefficients provide **very different numerical values** when applied to the same data (ten Hove et al., 2018)

Multilevel Structure and Reliability

Observational studies in social and behavioral science often have a multilevel structure. For example,

- subjects nested within clusters are rated by the same set of raters

$$Y_{i(k)j} = \mu + C_k + S_{i(k)} + R_j + (CR)_{kj} + \epsilon_{i(k)j}$$

- repeated measures for every subject that nested within clusters

$$Y_{j(i(k))} = \mu + C_k + S_{i(k)} + \epsilon_{j(i(k))}$$

Multilevel modeling is useful to distinguish or predict variation among clusters, and variation among subjects within clusters.

- **However, few studies define or report reliability specifically for different hierarchical levels** (Geldhof et al., 2014, ten Hove et al., 2022).

Multilevel Structure and Reliability

- Two recent papers (ten Hove et al., 2022; Schöenbrodt et al., 2022) address how to quantify reliability in multilevel settings based on generalizability theory.
- Nevertheless, both approaches give **inconsistent results** regarding their definition of cluster-level reliability.

Is there a general procedure to define reliability coefficients in various model structures?

Outline

Introduction

Generalizability Theory and the Correlation Approach

Reliability for Multilevel Data

Generalizability Theory (GT)

GT extends classical test theory by recognizing and estimating the magnitude of the **multiple sources of measurement error**, such as subjects, raters, items, occasions, etc. (Cronbach, 1972; Brennan, 2001).

- E.g., a three-way $Subject(S)\times Rater(R)\times Day(D)$ design

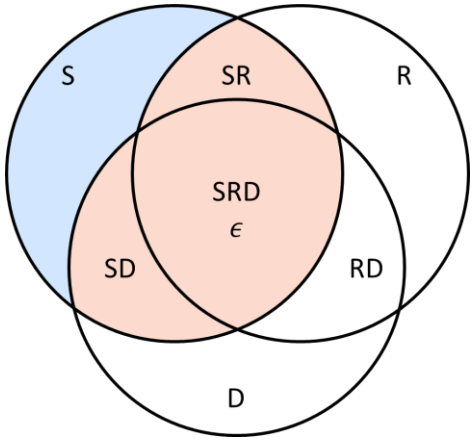
	Observer 1		Observer 2	
Patient	Day 1	Day 7	Day 1	Day 7
1	Y_{111}	Y_{112}	Y_{121}	Y_{122}
2	Y_{211}	Y_{212}	Y_{221}	Y_{222}
3	Y_{311}	Y_{312}	Y_{321}	Y_{322}
.	.	.	.	.
.	.	.	.	.
10	Y_{1011}	Y_{1012}	Y_{1021}	Y_{1022}

$$Y_{ijt} = \mu + S_i + R_j + D_t + (SR)_{ij} + (SD)_{it} + (RD)_{jt} + \epsilon_{ijt}$$

A $Subject(\mathrm{S}, i) \times Rater(R, j) \times Day(D, t)$ design

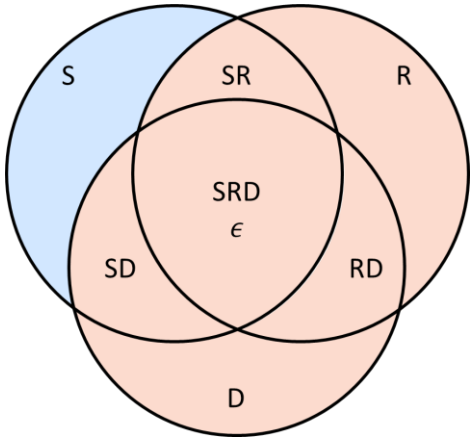
- *Generalizability coefficient* for relative decisions

$$\mathrm{GC} = \frac{\sigma^2_{\text{target}}}{\sigma^2_{\text{target}} + \sigma^2_{\text{rel.error}}} = \frac{\sigma^2_S}{\sigma^2_S + \sigma^2_{SR} + \sigma^2_{SD} + \sigma^2_\epsilon}$$



- *Dependability coefficient* for absolute decisions

$$\mathrm{DC} = \frac{\sigma^2_{\text{target}}}{\sigma^2_{\text{target}} + \sigma^2_{\text{abs.error}}} = \frac{\sigma^2_S}{\sigma^2_S + \sigma^2_R + \sigma^2_D + \sigma^2_{SR} + \sigma^2_{SD} + \sigma^2_{RD} + \sigma^2_\epsilon}$$



$\text{target} = E_{jt}[Y_{ijt}]$

$\text{rel. error} = (Y_{ijt} - E_{jt}[Y_{ijt}]) - (E_i[Y_{ijt}] - E[Y_{ijt}])$

$\text{abs. error} = Y_{ijt} - E_{jt}[Y_{ijt}]$

Correlation Approach

Vangeneugden et al. (2005) define

- the reliability coefficient as the correlation between two measurements from the same subject i, on different occasions/raters/items j and j'…

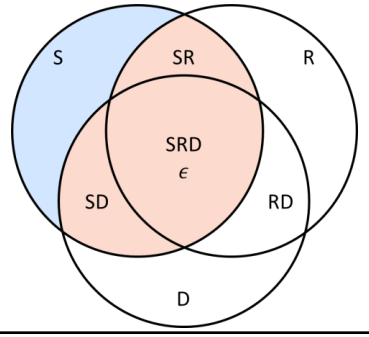
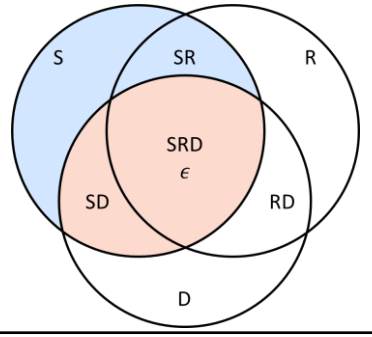
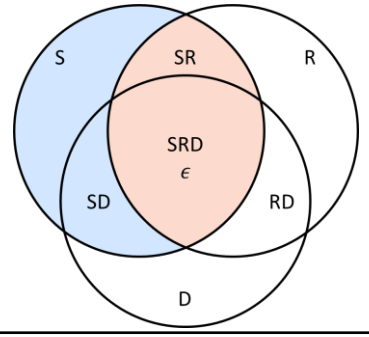
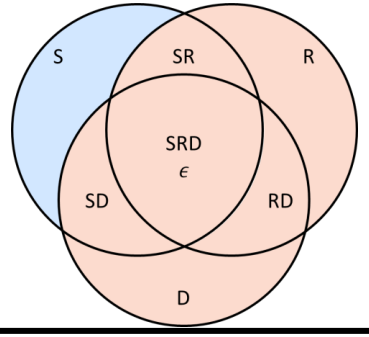
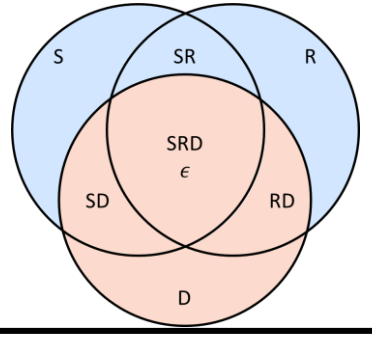
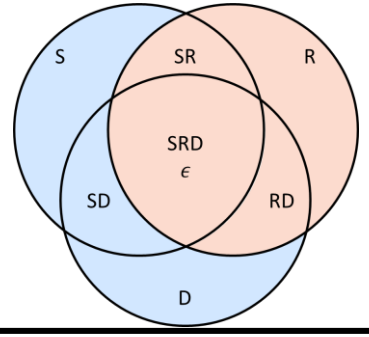
- Relative/Consistency reliability ← conditional correlation. E.g.,

$$Cor(Y_{ij}, Y_{ij'}|j, j')$$

  - keeping factors j and j' fixed
- Absolute/Agreement reliability ← (marginal) correlation . E.g.,

$$Cor(Y_{ij}, Y_{ij'})$$

GT vs. Correlation Approach

Reliability	Overall	Test-retest	Interrater
Relative Consistency	$\mathbf{GC} = \mathrm{Cor}(Y_{ijt}, Y_{ij't'} \mid j, j', t\ t') = \frac{\sigma_S^2}{\sigma_S^2 + \sigma_{SR}^2 + \sigma_{SD}^2 + \sigma_\epsilon^2}$ 	$\mathrm{Cor}(Y_{ijt}, Y_{ijt'} \mid j, t\ t') = \frac{\sigma_S^2 + \sigma_{SR}^2}{\sigma_S^2 + \sigma_{SR}^2 + \sigma_{SD}^2 + \sigma_\epsilon^2}$ 	$\mathrm{Cor}(Y_{ijt}, Y_{ij't} \mid j, j', t) = \frac{\sigma_S^2 + \sigma_{SD}^2}{\sigma_S^2 + \sigma_{SR}^2 + \sigma_{SD}^2 + \sigma_\epsilon^2}$ 
Absolute Agreement	$\mathbf{DC} = \mathrm{Cor}(Y_{ijt}, Y_{ij't'}) = \frac{\sigma_S^2}{\sigma_S^2 + \sigma_R^2 + \sigma_D^2 + \sigma_{SR}^2 + \sigma_{SD}^2 + \sigma_{RD}^2 + \sigma_\epsilon^2}$ 	$\mathrm{Cor}(Y_{ijt}, Y_{ijt'}) = \frac{\sigma_S^2 + \sigma_R^2 + \sigma_{SR}^2}{\sigma_S^2 + \sigma_R^2 + \sigma_D^2 + \sigma_{SR}^2 + \sigma_{SD}^2 + \sigma_{RD}^2 + \sigma_\epsilon^2}$ 	$\mathrm{Cor}(Y_{ijt}, Y_{ij't}) = \frac{\sigma_S^2 + \sigma_D^2 + \sigma_{SD}^2}{\sigma_S^2 + \sigma_R^2 + \sigma_D^2 + \sigma_{SR}^2 + \sigma_{SD}^2 + \sigma_{RD}^2 + \sigma_\epsilon^2}$ 

Outline

Introduction

Generalizability Theory and the Correlation Approach

Reliability for Multilevel Data

Comparison

	ten Hove et al. (2022)	Schönbrodt et al. (2022)
Variance decomposition	Generalizability theory	
Reliability coefficient	In terms of ICCs	Extend from Shrout & Land's (2012) formulas which are based on the DC
R for the subject level	Do not contain any cluster-related (higher level) variance components (i.e., σ_C^2, σ_{CR}^2)	
R for the cluster level	Do not contain any subject-related variance components (i.e., $\sigma_{S(C)}^2, \sigma_{S(C)R}^2$)	Contain subject-related variance components
Types	Separate *relative* and *absolute* reliability	Only discuss *absolute* reliability
Other characteristics	• Discuss subject-level and cluster-level reliability in three kinds of nested structure	• Discuss the roles of the *fixed effects* • Also build up the within-subject reliability under two different time scales

Two Nested Structures

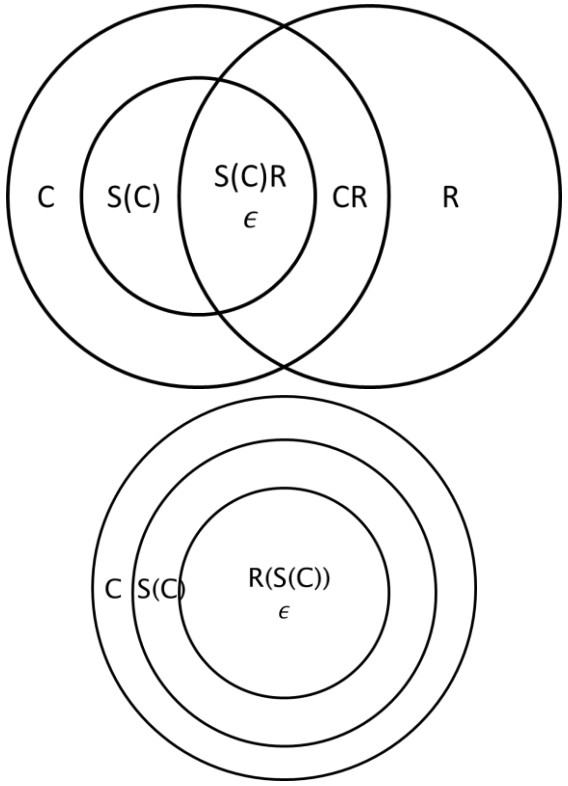
In the following cases, subjects (S) are nested within clusters (C), and …

- Case 1: raters (R) are *crossed with* S & C

$$Y_{i(k)j} = \mu + C_k + S_{i(k)} + R_j + (CR)_{kj} + \epsilon_{i(k)j}$$

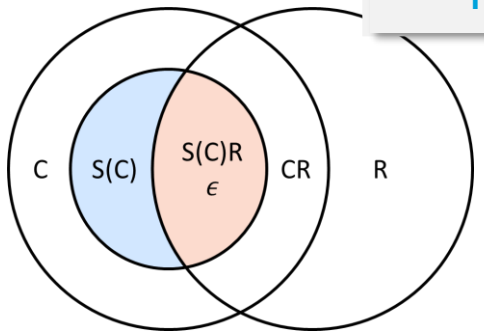
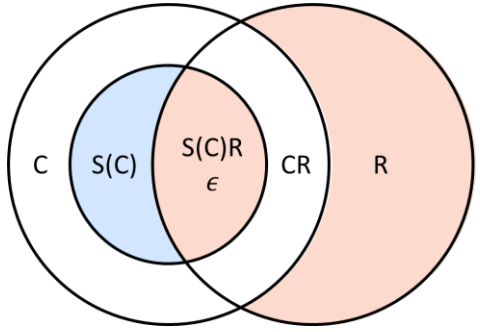
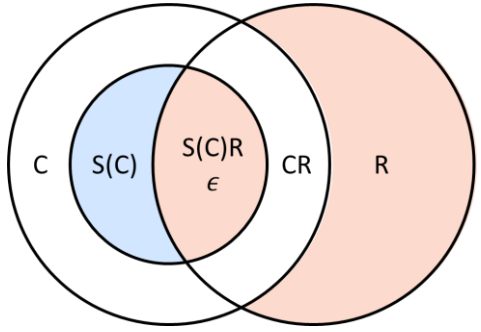
- Case 2: raters are *nested within* S & C

$$Y_{j(i(k))} = \mu + C_k + S_{i(k)} + \epsilon_{j(i(k))}$$

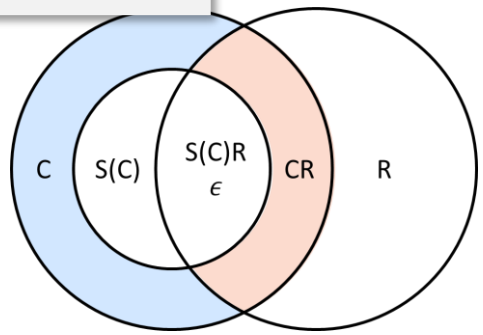
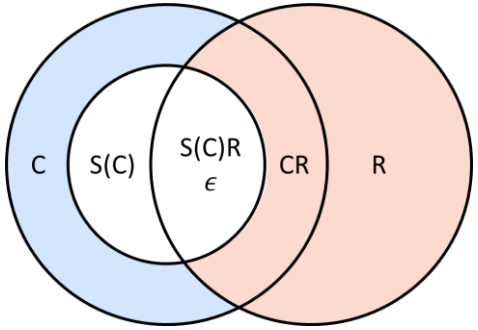
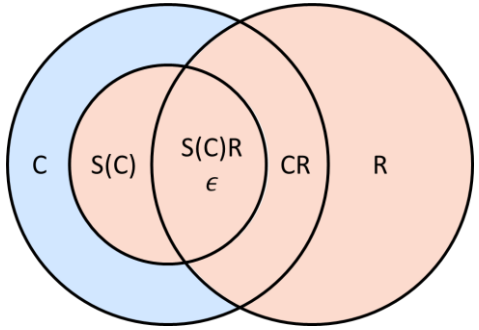


In here, we only consider the random effects model and compare the reliability for the single measurement.

Case 1: $Y_{i(k)j} = \mu + C_k + S_{i(k)} + R_j + (CR)_{kj} + \epsilon_{i(k)j}$

	ten Hove et al. (2022)	Schönbrodt et al. (2022)	Correlation approach
Subject-level	$\frac{\sigma_S^2}{\sigma_S^2 + \sigma_\epsilon^2}$ 	Ignore all higher-level variances	$\mathrm{Cor}(Y_{i(k)j}, Y_{i(k)j'} \mid k, j, j') = \frac{\sigma_S^2}{\sigma_S^2 + \sigma_\epsilon^2}$
R^S_{rel}		(not defined)	
R^S_{abs}	$\frac{\sigma_S^2}{\sigma_S^2 + \sigma_R^2 + \sigma_\epsilon^2}$ 	$\frac{\sigma_S^2}{\sigma_S^2 + \sigma_R^2 + \sigma_\epsilon^2}$ 	$\mathrm{Cor}(Y_{i(k)j}, Y_{i(k)j'} \mid k) = \frac{\sigma_S^2}{\sigma_S^2 + \sigma_R^2 + \sigma_{CR}^2 + \sigma_\epsilon^2}$

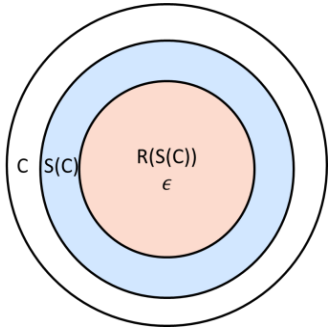
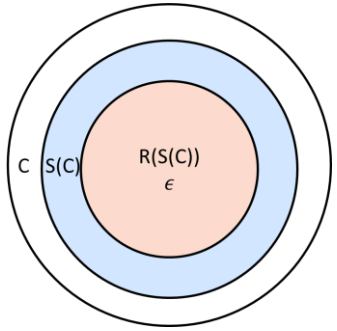
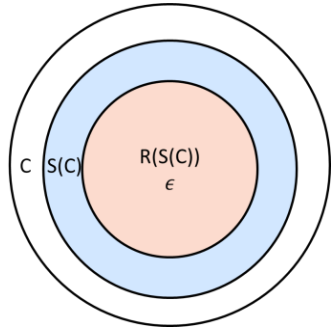
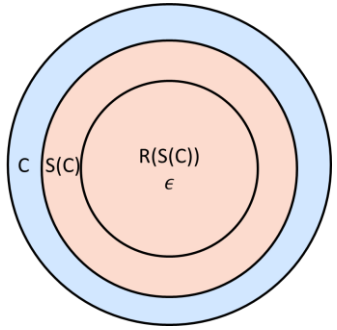
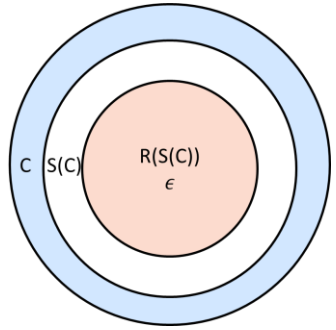
Case 1: $Y_{i(k)j} = \mu + C_k + S_{i(k)} + R_j + (CR)_{kj} + \epsilon_{i(k)j}$

	ten Hove et al. (2022)	Schönbrodt et al. (2022)	Correlation approach
R^C_{rel}	$\frac{\sigma_C^2}{\sigma_C^2 + \sigma_{CR}^2}$ 	(not defined)	$\mathrm{Cor}(Y_{i(k)j}, Y_{i(k)j'} \mid i_{(k)}, j, j') = \frac{\sigma_C^2}{\sigma_C^2 + \sigma_{CR}^2 + \sigma_\epsilon^2}$
R^C_{abs}	$\frac{\sigma_C^2}{\sigma_C^2 + \sigma_R^2 + \sigma_{CR}^2}$ 	$\frac{\sigma_C^2}{\sigma_C^2 + \sigma_S^2 + \sigma_R^2 + \sigma_{CR}^2 + \sigma_\epsilon^2}$ 	$\mathrm{Cor}(Y_{i(k)j}, Y_{i(k)j'} \mid i_{(k)}) = \frac{\sigma_C^2}{\sigma_C^2 + \sigma_R^2 + \sigma_{CR}^2 + \sigma_\epsilon^2}$

Ignore all lower-level variances

$= \mathrm{Cor}(Y_{i(k)j}, Y_{i'(k)j'})$

Case 2: $Y_{j(i(k))} = \mu + C_k + S_{i(k)} + \epsilon_{j(i(k))}$

	ten Hove et al. (2022)	Schönbrodt et al. (2022)*	Correlation approach
Subject-level	$\frac{\sigma_S^2}{\sigma_S^2 + \sigma_\epsilon^2}$	$\frac{\sigma_S^2}{\sigma_S^2 + \sigma_\epsilon^2}$	$\mathrm{Cor}\left(Y_{i(k)j}, Y_{i(k)j'} \mid k \right) = \frac{\sigma_S^2}{\sigma_S^2 + \sigma_\epsilon^2}$
R^S			
Cluster-level		$\frac{\sigma_C^2}{\sigma_C^2 + \sigma_S^2 + \sigma_\epsilon^2}$	$\mathrm{Cor}\left(Y_{i(k)j}, Y_{i(k)j'} \mid i_{(k)} \right) = \frac{\sigma_C^2}{\sigma_C^2 + \sigma_\epsilon^2}$
R^C	Cannot be defined because it ignores all lower-level variances		

* See also Raudenbush et al. (1991), Cranford et al. (2006), and Shrout & Lane (2012).

Discussion and Future Work

- Point out the different reliability definitions for multilevel data in the latest literature (ten Hoev et al., 2022; Schönbrodt et al., 2022).
  - there is no consensus on defining the reliability coefficient or telling which one is better.
- Using correlation to define reliability for multilevel data may be a promising approach
  - Easy to understand which kind of reliability from the formula.
  - Has the flexibility for more complicated models, e.g., (generalized) linear mixed-effects models with serial correlation (Vangeneugden et al., 2005; Molenberghs et al. 2007)
- As for estimation, we will apply a Bayesian framework with Markov chain Monte Carlo (MCMC) to estimate reliability coefficients.

References

- Brennan, R. L. (2001). *Generalizability theory* (1st ed.). Springer New York. https://doi.org/10.1007/978-1-4757-3456-0
- Cranford, J. A., Shrout, P. E., Iida, M., Rafaeli, E., Yip, T., & Bolger, N. (2006). A procedure for evaluating sensitivity to within-person change: Can mood measures in diary studies detect change reliably? *Personality & Social Psychology Bulletin*, *32*(7), 917–929. https://doi.org/10.1177/0146167206287721
- Cronbach, L. J. (1972). The dependability of behavioral measurements. *Theory of Generalizability for Scores and Profiles*, 1–33.
- Geldhof, G. J., Preacher, K. J., & Zyphur, M. J. (2014). Reliability estimation in a multilevel confirmatory factor analysis framework. *Psychological Methods*, *19*, 72–91. https://doi.org/10.1037/a0032138
- Lord, F. M., & Novick, M. R. (2008). *Statistical theories of mental test scores*. IAP.
- Molenberghs, G., Laenen, A., & Vangeneugden, T. (2007). Estimating reliability and generalizability from hierarchical biomedical data. *Journal of Biopharmaceutical Statistics*, *17*(4), 595–627. https://doi.org/10.1080/10543400701329448
- Schönbrodt, F. D., Zygar-Hoffmann, C., Nestler, S., Pusch, S., & Hagemeyer, B. (2022). Measuring motivational relationship processes in experience sampling: A reliability model for moments, days, and persons nested in couples. *Behavior Research Methods*, *54*(4), 1869–1888. https://doi.org/10.3758/s13428-021-01701-7

References

- Shavelson, R. J., Webb, N. M., & Rowley, G. L. (1989). Generalizability theory. *American Psychologist*, *44*(6), 922.
- Shrout, P. E., & Fleiss, J. L. (1979). Intraclass correlations: Uses in assessing rater reliability. *Psychological Bulletin*, *86*(2), 420–428. https://doi.org/10.1037/0033-2909.86.2.420
- Shrout, P. E., & Lane, S. P. (2012). Psychometrics. In M. R. Mehl & T. S. Conner (Eds.), *Handbook of research methods for studying daily life* (pp. 302–320). The Guilford Press. https://psycnet.apa.org/record/2012-05165-017
- ten Hove, D., Jorgensen, T. D., & van der Ark, L. A. (2018). On the usefulness of interrater reliability coefficients. In M. Wiberg, S. Culpepper, R. Janssen, J. González, & D. Molenaar (Eds.), *Quantitative Psychology: The 82nd Annual Meeting of the Psychometric Society, Zurich, Switzerland, 2017* (pp. 67-75). (Springer Proceedings in Mathematics & Statistics; Vol. 233). Springer. https://doi.org/10.1007/978-3-319-77249-3_6
- ten Hove, D., Jorgensen, T. D., & van der Ark, L. A. (2022). Interrater reliability for multilevel data: A generalizability theory approach. *Psychological Methods*, *27*(4), 650–666. https://doi.org/10.1037/met0000391
- Vangeneugden, T., Laenen, A., Geys, H., Renard, D., & Molenberghs, G. (2005). Applying concepts of generalizability theory on clinical trial data to investigate sources of variation and their impact on reliability. *Biometrics*, *61*(1), 295–304. https://doi.org/10.1111/j.0006-341X.2005.031040.x

Q&A

Reliability for multilevel data: A correlation approach

PhD candidate: Tzu-Yao Lin[1,2]

Supervisors: Dr. Francis Tuerlinckx[2], Dr. Sophie Vanbelle[1]

[1]Department of Methodology and Statistics, Care and Public Health Research Institute, Maastricht University, the Netherlands

[2]Quantitative Psychology and Individual Differences Research Group, Faculty of Psychology and Educational Sciences, KU Leuven, Belgium

Maastricht University KU LEUVEN

20-06-2023